

DIABETIC RETINOPATHY SCREENING REPORT

Smart India Hackathon 2026 | Problem 26038 | MathWorks Telemedicine Pipeline

Patient / Image: 4a693dd3921a

Screening Date: 28-Sep-2026 03:51

Quality: **UNGRADEABLE (1/3)**

AI CLINICAL DIAGNOSIS

Proliferative DR

ICDR Severity Grade: Level 4 of 4

IMMEDIATE SPECIALIST REFERRAL — Sight-threatening retinopathy

KEY CLINICAL LESION EVIDENCE

Microaneurysms (MA)

Vascular microleaks

66 detected

Hard Exudates (EX)

Lipid deposits

12 detected

Hemorrhages (HE)

Vascular ruptures

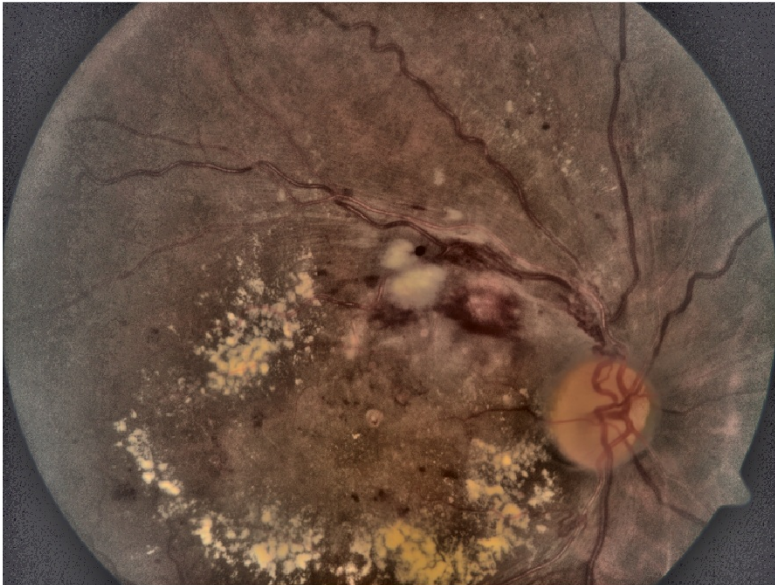
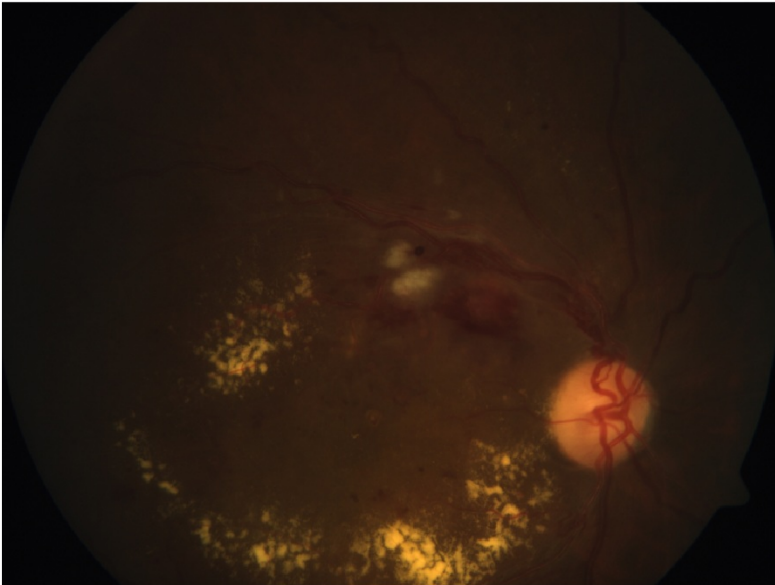
0 detected

Optic Disc Center: [X: 593, Y: 412] | Fovea Center: [X: 510, Y: 417]

FUNDUS IMAGE ACQUISITION & ADAPTIVE ENHANCEMENT

Original Fundus Photograph

CLAHE Enhanced & Illumination Normalized



RETINAL SEGMENTATION & EXPLAINABLE AI

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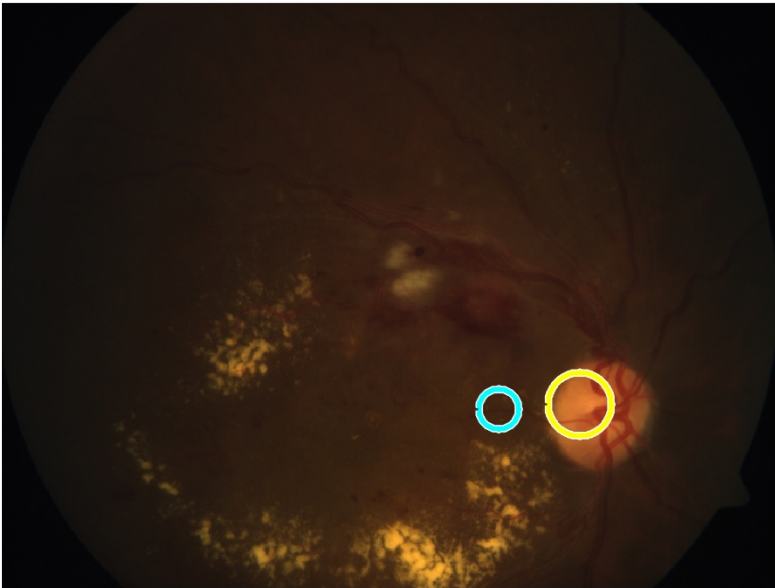
Patient / Image: 4a693dd3921a

AI Diagnosis: Proliferative DR

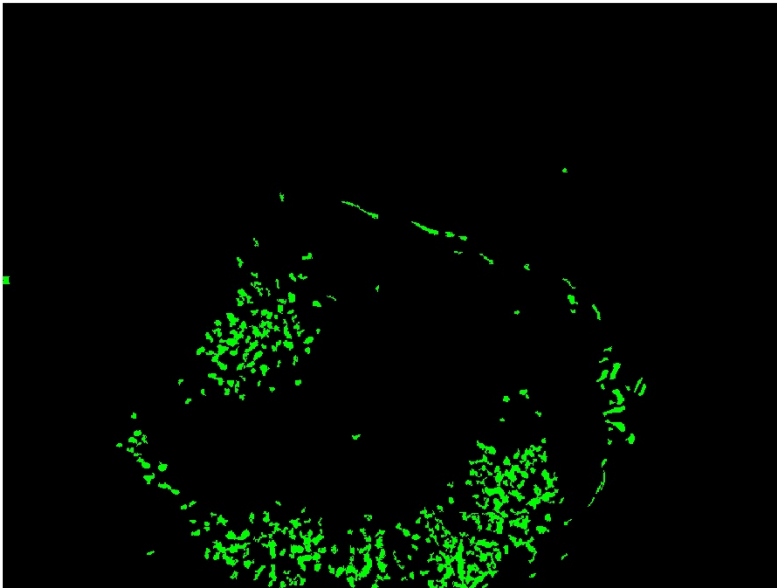
Action: CRITICAL REFERRAL (Proliferative DR)

4-AXIS CLINICAL VALIDATION DASHBOARD

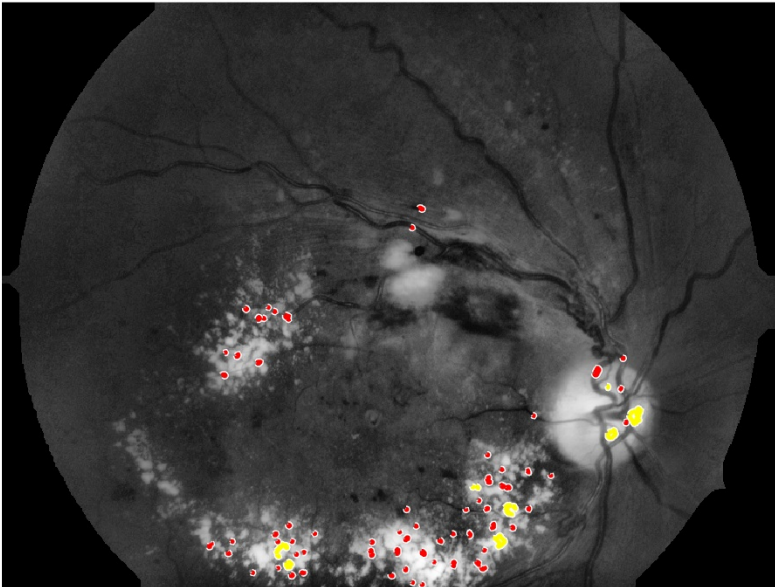
1. Landmarks (Optic Disc=Yellow, Fovea=Cyan)



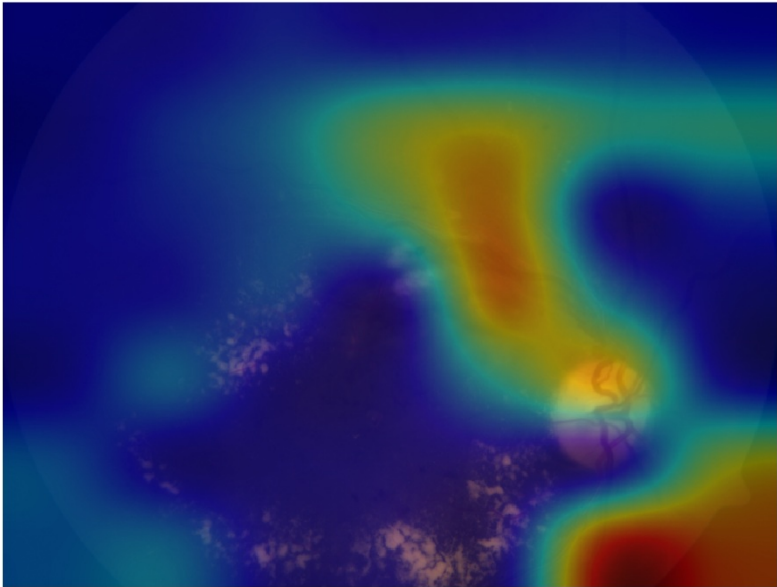
2. Retinal Vascular Network Segmentation



3. Verified Lesions (MA=66, EX=12, HE=0)



4. Grad-CAM Explainable AI Attention Heatmap



CLINICAL ACTION: CRITICAL REFERRAL (Proliferative DR)

Recommendation : IMMEDIATE SPECIALIST REFERRAL — Sight-threatening retinopathy

Telemedicine Protocol: Ophthalmologist validates findings in < 30 seconds via Grad-CAM attention overlay.